BTEC Level 3 Applied Science · Unit 1 Physics · Worksheet

Progressive Waves

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. Define amplitude, wavelength and frequency.
2. Write the wave equation and give SI units for each quantity.
3. How are period and frequency related?

B Knowledge

4. What does a progressive wave transfer, and what does it not transfer?
5. Which quantity stays the same when a wave enters a new medium?
6. How is amplitude read from a displacement–distance graph?

C Calculations

7. $f = 4.0$ Hz, $\lambda = 0.80$ m. Calculate v.
8. A radio wave has $f = 200$ MHz. Take $c = 3.00 \times 10^8$ m s^{-1}. Calculate λ.
9. A wave travels 48 m in 0.16 s. If $\lambda = 2.0$ m, calculate f.

D Exam-style practice

10. Define amplitude and wavelength. [2]
11. A note of 510 Hz travels at 340 m s^{-1}. Calculate λ. [2]
12. Explain why air particles do not travel from a loudspeaker to a listener. [3]

E Challenge

13. A displacement–time trace has 5 complete cycles in 20 ms. The matching snapshot shows 4.0 cm between adjacent crests. Calculate f, T, λ and v. Show unit conversions.